

respectively. Two external pull-up resistors R9 and R10 are used along with the switches for changing the display mode of the supply and for enabling/disabling the serial transmission. An LED (LED2) is connected to port pin PC3, which is used to display the status of serial transmission. A piezobuzzer is also connected to port pin PB0, which gives audio beeps when the device is switched on or a button/switch is pressed to change the menu.

As shown in the circuit, the lower 'D' port is connected to the lower nibble of the LCD, and EN and RS pins on the LCD are connected to PD3 and PD2 of IC1, respectively. Pin R/W on the LCD is grounded. Preset VR1 (10-kilo-ohm) is used to adjust the contrast of the LCD.

TXD and RXD pins of the microcontroller are interfaced with the MAX232 on pins 11 and 7. Pins 10

and 14 are connected to the external DB9 serial connector, which can be connected to the computer via a serial cable. Pins 16 and 15 of MAX232 are  $V_{CC}$  and ground, respectively.

Pins 20, 21 and 22 on the ATmega8 set up the configuration for using internal 2.56V voltage reference for the in-built ADC of the chip. Pin 1, which is the reset pin, is pulled up with the help of an external 10-kilo-ohm resistor. Reset switch S1 can be used to reset the microcontroller.

A voltage divider made up of resistors R6 and R5 scales down the variable supply channel voltage so that it can be interfaced to the in-built ADC of ATmega8. This is connected to ADC1 (pin PC1) of the controller.

For current sensing we use a shunt resistor of 0.3 ohms in series with the variable supply channel. Thus a voltage ( $V_{sense}$ ) develops across the

resistor, which is amplified with the help of non-inverting amplifier 741 (op-amp). The gain is set to 1.7 with the help of resistors R4 and R7.

Connection point 'A' is connected to the shunt resistor, and connection point 'B' to the positive terminal

of LM338 output as shown in the block diagram. Shunt resistor should be connected externally at Point A on the PCB of Fig. 10. The ground of the two sections is made common as it is necessary to interconnect the ground of both the sections.

## Software program

The code for the microcontroller is developed using BASIC language in BASCOM compiler. Initially, we set the microcontroller configuration file and the crystal frequency. It is important to note that the code is developed to run on 4MHz frequency and the internal 4MHz R-C oscillator of the microcontroller is used here; this frequency is more than enough for our operations. The baud rate is set to 9600. After that, we initialise various variables to implement the algorithm, and the ADC, I/O ports, LCD are also configured.

The software uses two functions, *Mavgcurrent* and *Mavgvoltage*, to implement the moving average window for the calculated values of current and voltage. This increases the accuracy of calculations and also maintains stability of the display variables. Instantaneous power is also calculated and displayed using these variables only.

After getting readings from ADC0 and ADC1, the software calculates the current and voltage values by using inverse Ohm's law, and displays the values on the LCD. For serial transmission, it takes the values of current, voltage and power in a string and sends the string to a computer that is running a



Fig. 6: The graphical user interface on the computer monitor

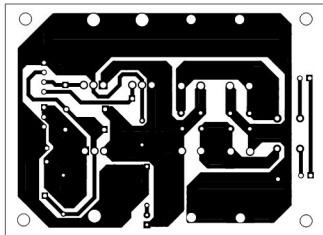


Fig. 7: An actual-size, single-side PCB for power supply section

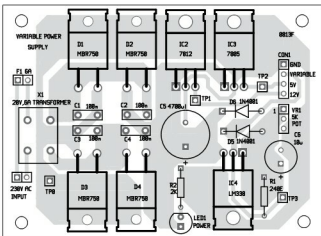


Fig. 8: Component layout for the PCB in Fig. 7